

Exploratory Data Analysis (EDA) Report – Diwali Sales Data

Objective

The purpose of this analysis is to explore customer purchase patterns from the Diwali Sales Dataset to identify trends in buying behavior across different demographics such as gender, age, state, and marital status. The results aim to support data-driven business decisions for targeted marketing and sales strategies.

1. Data Preparation

- Libraries Used: NumPy, Pandas, Matplotlib, Seaborn
- Dataset: Diwali Sales Data.csv
- Steps:
 - Loaded dataset and inspected structure
 - Dropped irrelevant columns ('Status', 'unnamed1')
 - Removed missing values
 - Converted 'Amount' column to integer type

2. Univariate and Bivariate Analysis

****Gender Analysis:**** Female buyers dominate both in count and total spending (Count Plot, Bar Plot)

****Age Group:**** Most buyers are aged 26–35 years; majority are females (Count Plot with Gender Hue)

****State-Wise:**** Top 3 states: Uttar Pradesh, Maharashtra, Karnataka (Bar Plot)

****Marital Status:**** Married women have highest purchasing power (Count Plot, Bar Plot)

****Occupation:**** Highest spending observed among IT, Healthcare, and Aviation professionals (Bar Plot)

3. Key Insights

1. Female buyers contribute the most to total sales.
2. Age group 26–35 is the most active buying segment.
3. Married women have higher purchasing power.
4. Top-performing states: Uttar Pradesh, Maharashtra, Karnataka.
5. IT and Healthcare professionals spend the most.

4. Business Recommendations

- Focus marketing on married women aged 26–35 years.
- Offer personalized festive discounts.
- Prioritize regional promotions in Uttar Pradesh, Maharashtra, and Karnataka.
- Develop premium products for IT and healthcare professionals.

5. Conclusion

The analysis reveals that gender, age, marital status, and region strongly influence purchasing behavior. Businesses can use these insights to design targeted marketing strategies and boost festive sales performance.